
LIRAF

Lunar Integrated Resource Assessment Framework

An Integrated Multi-Source Framework for Lunar Ice Detection, Landing Site Selection, Rover Traverse Planning, and Resource Assessment

Submitted By

Team Name: LunaForge

Team Size: 4 Members

Challenge 8

Detection and Characterization of Subsurface Ice in Lunar South Polar Regions Using Chandrayaan-2 Radar and Imagery Data for Landing Site and Rover Traverse Planning

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LIRAF – Detection and Characterization of Subsurface Ice in Lunar South Polar Regions

An end-to-end decision-support framework for subsurface ice characterization and mission planning in the lunar south polar region.

1. Problem Understanding

The lunar south pole contains Permanently Shadowed Regions (PSRs) where water ice may have remained preserved for billions of years. This resource is vital for In-Situ Resource Utilization (ISRU), supporting oxygen production, life support, and propellant generation for future lunar missions. Accurate identification of accessible ice-bearing regions is therefore a key objective of lunar exploration.

Current lunar datasets provide complementary but incomplete information. Radar reveals subsurface scattering, hydrogen data indicates volatile enrichment, DEMs describe terrain, and illumination analysis identifies PSRs. However, analysing these datasets independently results in uncertain ice detection and overlooks operational constraints such as landing safety and rover traversability.

To address these limitations, LIRAF integrates radar, terrain, hydrogen, and illumination data into a unified decision-support framework for ice detection, landing site selection, rover traverse planning, and scenario-based resource assessment.

2. Proposed Solution

Core Approach

LIRAF integrates radar observations, terrain analysis, hydrogen abundance, and illumination modelling into a unified framework for detecting potential subsurface ice while supporting landing site selection, rover navigation, and preliminary resource assessment. Instead of relying on a single dataset, complementary observations are fused to improve detection confidence and operational feasibility.

Methodology

Module	Description	Dataset(s) Used	Status
Radar Processing	Generates CPR and DOP maps from dual-frequency polarimetric SAR data, identifies candidate ice pixels using literature-based thresholds ($CPR > 1$, $DOP < 0.13$) with a crater-ejecta false-positive filter, and uses OHRC imagery to assess crater morphology, boulder distribution, and landing-site safety around candidate PSRs, providing surface context rather than direct subsurface ice validation.	LRO Mini-RF (proxy, pre-finale) → Chandrayaan-2 DFSAR (target dataset at finale)	Pipeline validated using Mini-RF proxy data; adapted to DFSAR at Hour 0 through preprocessing and parameter calibration.
Terrain Analysis	Processes LOLA DEM to generate elevation, slope, hillshade, illumination, and PSR maps. Illumination modelling follows the methodology of Mazarico et al.	LOLA DEM + SPICE kernels (used	Fully implemented and validated.

	(2011), adapted using 72 azimuth samples (10° intervals) for computational efficiency.	pre-finale and at finale)	
Data Fusion	Integrates radar, hydrogen abundance, terrain, and illumination datasets using GMM-based clustering and spatial interpolation to identify high-confidence exploration targets. GMM clustering was selected because it models overlapping feature distributions better than hard clustering methods, improving confidence in candidate ice region identification.	Outputs of Radar Processing + Terrain Analysis + Lunar Prospector	Fusion logic validated on Mini-RF-derived outputs; re-runs automatically on DFSAR-derived outputs at finale.
Mission Planning	<i>Ranks candidate landing sites using ice confidence, slope, illumination, terrain safety, and rover accessibility, generates A*/RRT rover traverse paths, and estimates subsurface ice volume using dielectric mixing models. A*/RRT computes optimal paths on structured terrain, while RRT provides feasible alternatives in complex obstacle-rich environments.</i>	Outputs of Data Fusion + LOLA DEM	Pipeline validated using Mini-RF proxy data; final ice volume will be computed using the DFSAR-derived ice confidence map during the finale.

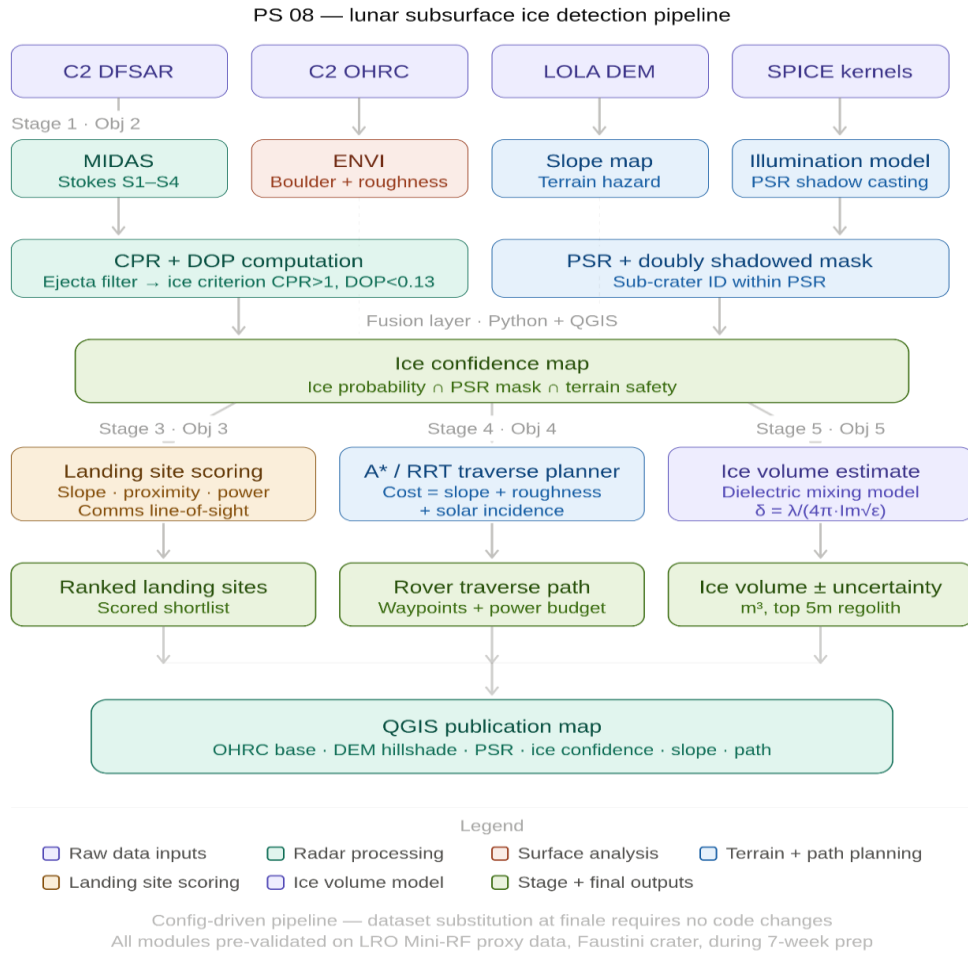


Figure 1. Overall LIRAF processing pipeline

Key Innovations

- Multi-source data fusion: Integrates radar polarimetry, hydrogen abundance, terrain morphology, and illumination modelling into a single decision-support framework rather than relying on any one dataset's incomplete picture. LIRAF cross-validates ice detection across independent physical signatures.
- Dual-criterion ice detection: Applies a combined CPR + DOP threshold ($CPR > 1$, $DOP < 0.13$) with a crater-ejecta false-positive filter, directly addressing the well-documented radar ambiguity between ice-bearing terrain and rough rocky surfaces.
- Validated-then-substituted pipeline: All four modules are built and validated in advance using LRO Mini-RF as a structural proxy for Chandrayaan-2 DFSAR. During the finale, the team executes a rehearsed and debugged pipeline on the released DFSAR data rather than developing the methodology under 30-hour time constraints.
- Operations-complete mission planning: Extends beyond ice detection by producing a ranked landing-site shortlist, A^*/RRT optimized rover traverse paths* with slope and solar-power profiling, and quantitative ice-volume estimates, providing a complete decision-support package rather than only a detection result.
- Config-driven architecture: Modular processing pipeline enabling rapid adaptation to future Chandrayaan and lunar mission datasets with minimal code changes.

Data Sources

Dataset	Purpose	Role	
Chandrayaan-2 DFSAR	CPR and DOP Computation	Target Dataset – supplied by ISRO at finale	
OHRC (Orbiter High Resolution Camera)	Surface morphology validation, boulder detection, and landing site assessment	Supports landing-site safety assessment and surface context validation around candidate PSRs; due to limited illumination inside PSRs, OHRC is used for contextual interpretation rather than direct subsurface ice validation.	
Lunar Prospector	Hydrogen abundance	Supplementary cross-validation, publicly available	
LOLA DEM	Terrain Analysis	Publicly available – used both pre-finale and at finale	
Illumination Model	Terrain Analysis	Derived from DEM + SPICE – used both pre-finale and at finale	
LRO Mini-RF	CPR and DOP Computation	Proxy dataset – used for pre-finale pipeline validation	

The proposed framework utilizes publicly available planetary datasets from NASA while remaining compatible with the objectives of the BAH Challenge based on Chandrayaan-2 observations.

3. Feasibility & Implementation Plan

Implementation Readiness

Core modules have been implemented using publicly available planetary datasets. The remaining work focuses on integration, optimization, and final demonstration during the 30-hour hackathon finale. If direct DFSAR integration requires additional preprocessing, the validated Mini-RF pipeline will be used to demonstrate the complete workflow while adapting the final stage to the provided DFSAR dataset.

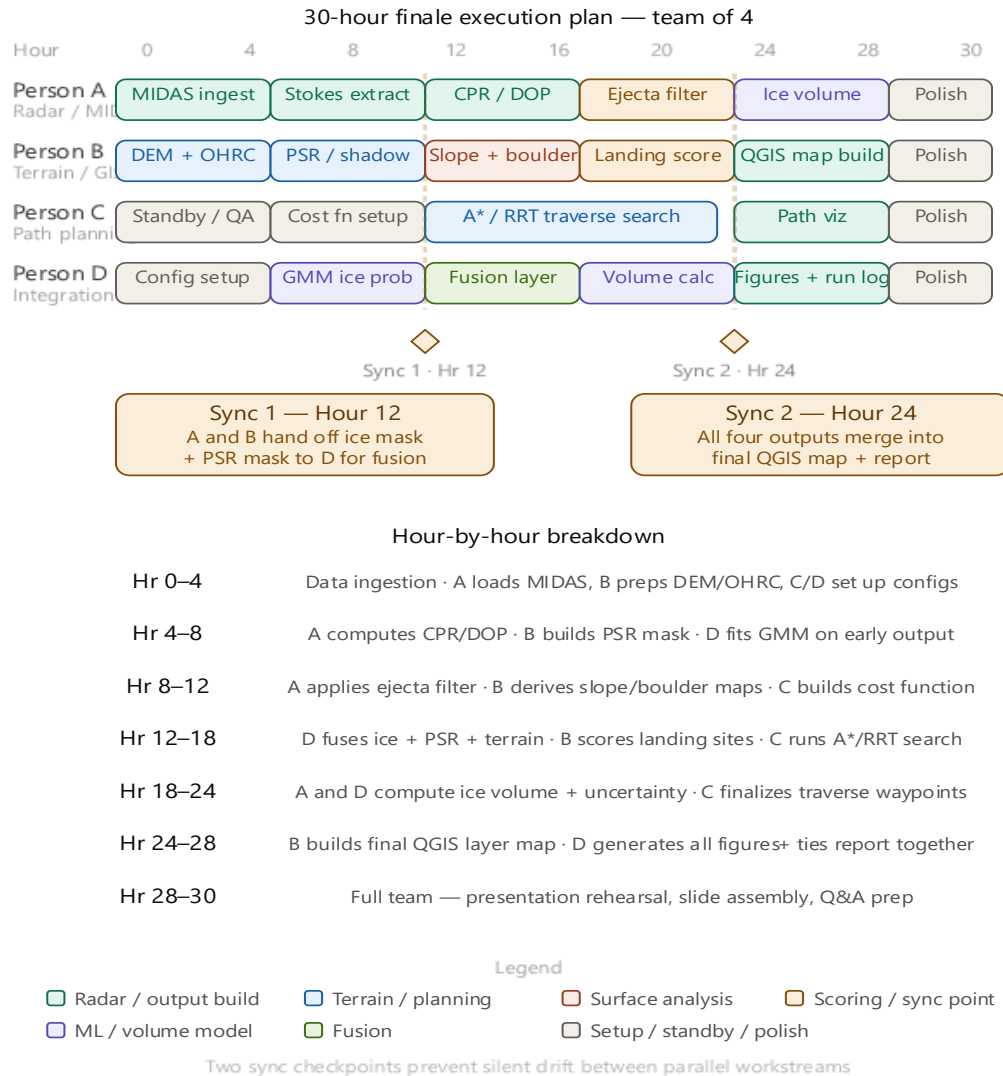


Figure 2. Proposed 30-hour implementation schedule for the BAH finale.

Technology Stack: Python • GDAL • Rasterio • QGIS • NumPy • Pandas • Matplotlib • Scikit-learn • CuPy • CUDA • cuML

4. Expected Impact

Expected Deliverables: The proposed framework supports scientific ice characterization and mission planning by integrating radar, terrain, illumination, and hydrogen information into a unified decision-support system. The expected outputs are summarized below.

Scientific Deliverables	Operational Deliverables
CPR & DOP Maps	Landing Site Recommendations

Hydrogen Abundance Map	Rover Traverse Paths
PSR & Terrain Maps	Ice Confidence Map
Candidate Ice Regions	Ice Volume & Mass Estimates

5. Team Expertise

Member	Role	Key Contribution	Primary Literature
Anjali Saini	Data Engineer & GIS	QGIS terrain analysis, DEM processing, PSR mapping	Mazarico et al. (2011)
Aditya Narayan	Remote Sensing	Radar processing (MIDAS), CPR/DOP computation, hydrogen analysis	Spudis et al. (2013)
Prince	AI/ML Engineer	Data fusion, clustering, A*/RRT path planning	Campbell et al. (2006)
Himanshu	Resource Analyst	Ice confidence, volume estimation, dashboard	Paige et al. (2010)

6. References

1. *NASA Planetary Data System (PDS). Lunar Reconnaissance Orbiter (LRO) Mini-RF Data Archive.*
2. *NASA Planetary Data System (PDS). Lunar Orbiter Laser Altimeter (LOLA) Digital Elevation Model.*
3. *Lunar Prospector Mission. Neutron Spectrometer Hydrogen Abundance Dataset.*
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5. **Spudis, P. D., Bussey, D. B. J., Baloga, S. M., et al. (2013).** Evidence for Water Ice on the Moon: Results from Mini-RF Radar Observations.
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9. **Paige, D. A., Siegler, M. A., Zhang, J. A., et al. (2010).** Diviner Lunar Radiometer Observations of Cold Traps in the Moon's Polar Regions.
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11. *ISRO. Bharatiya Antariksh Hackathon (BAH) 2026 – Challenge Statement.*